THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 23, 2022 TIME: 9:30 AM – 12:30 PM

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:
Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

IV. Ideal gas law constant = 0.082056 L·atm/(K·mol)
Partial pressure of CO_2 in atmosphere: $1 \times 10^{-3.5}$ atm
Henry's Law constant for CO_2: $1 \times 10^{-1.5}$ M/atm
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Henry's Law coefficient for O_2 (25 °C) = 0.001263 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1 \times 10^{-14}$

Notes:

This is a closed book examination. Internet searching and crowdsourcing from group messages, online forums or social media, etc. are strictly forbidden.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (Description and simple calculation questions with 3 marks each)

1. Carbon dioxide (CO_2) in ambient air is approximately 316 ppmv. Express this CO_2 concentration in mg/m^3 at 1 atm and 25 °C. (Hint: the molecular weight of CO_2 is 44 g/mol).
2. Chlorine is applied to disinfect drinkable water with an initial concentration of 1 mg/L. Considering the disinfection process as a batch reactor with a first-order decay coefficient of 0.05 hr^{-1}, how many hours does it take to reduce its concentration to 0.001 mg/L?
3. A wastewater contains 200 mg/L C_6H_7N. Determine its total oxygen demand (TBOD) and nitrogenous oxygen demand (NBOD). (Hint: the molecular weight of C_6H_7N is 93 g/mol)
4. List the three major categories of water pollutants and give one example for each category.
5. Some waste has a five-day BOD at 20 °C equal to 210 mg/L and an ultimate BOD of 350 mg/L. Find the five-day BOD at 25 °C. (Hint: θ is 1.047 for temperature correction of BOD_5)
6. What are the major chemical reactions for disinfection using Cl_2 gas? Compare the disinfection power of different chlorine species generated during Cl_2 gas disinfection.
7. What are the underlying assumptions when the rate of biomass growth and rate of biomass death are assumed to be 1st order kinetics?
8. List three purposes of wastewater treatment.
9. List the criteria air pollutants.
10. What are the major effects of climate change on ocean?

Section B. (Three questions with unequal marks)

11. (a) Estimate the pH of unpolluted rainwater. It is assumed that rainwater has reached chemical equilibrium with CO_2 in atmosphere.

(10 marks)

(b) Due to severe air pollution, the rainwater of a city contains 2 mM HNO_3. Determine the pH of the polluted rainwater (Hints: HNO_3 is a strong acid. You also need to set up a proper charge balance).

(12 marks)

12. (a) A mixture consisting of 30 mL of waste and 270 mL of seeded dilution water has an initial DO of 8.55 mg/L; after five days, it has a final DO of 2.40 mg/L. Another bottle containing just the seeded dilution water has an initial DO of 8.75 mg/L and a final DO of 8.53 mg/L. Find the five-day BOD of the waste.

(8 marks)

(b) A sample of water at pH 10.5 has 39.0 mg/L CO_3^{2-} and 24.5 mg/L HCO_3^-, please find its alkalinity as mg/L $CaCO_3$.

(10 marks)

(c) Air pollution is a global concerned issue, which greatly threatens public health and social development. Briefly discuss the origin of air pollutants and their control measures (**< 300 words**).

(10 marks)

13. You are an engineer working with the city's drainage services department. You have been tasked to design a sewage treatment plant with secondary treatment and sludge treatment facilities. You have chosen to adopt the activated sludge process and anaerobic digestion process. Answer the following questions:

(a) What are the **three similarities** and **three differences** between the activated sludge process and anaerobic digestion process.

(6 marks)

(b) Draw a flow diagram to illustrate a sewage treatment plant design with the activated sludge process (ASP) and anaerobic digestion (AD) process. On the diagram, denote the following mass flows for water (Q) and activated sludge biomass concentration (X):

(i) Q_i: flowrate of influent
(ii) Q_e: flowrate of secondary effluent
(iii) Q_w: flowrate of waste sludge
(iv) Q_r: flowrate of return sludge
(v) X_i: biomass concentration in influent
(vi) X: biomass concentration in the ASP bioreactor
(vii) X_r: biomass concentration in return sludge
(viii) X_e: biomass concentration in secondary effluent
(ix) V_r: volume of bioreactor

(6 marks)

(c) The activated sludge process treats 15,000 $m^3.d^{-1}$ of sewage and is operated at a hydraulic retention time of 12 hours. Calculate the volume of the ASP bioreactor.

(3 marks)

(d) Following the secondary treatment, 12,000 $m^3.d^{-1}$ of secondary effluent is discharged while the flowrates of the waste sludge and return sludge are the same. The activated sludge biomass concentration is 2,000 $mg.L^{-1}$ in the ASP bioreactor, 4,500 $mg.L^{-1}$ in the waste sludge and negligible in both the influent and secondary effluent. Calculate the solid retention time (SRT) for the activated sludge process (round your answer to the nearest hour).

Given:

$$SRT = \frac{V_r X}{Q_e X_e + Q_w X_r}$$

(5 marks)

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